

Interlocked Impact of Geometry and Coupling Strength on Maximal Mutual Information in a Nerve Tract

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Abstract

This paper quantifies how geometric configuration and ephaptic coupling jointly constrain mutual information in nerve tracts. Using Hodgkin-Huxley simulations and information-theoretic analysis, we show that synchronization is only feasible when axon inclinations remain below 45 degrees, and that increasing coupling strength reduces maximal mutual information due to inter-channel correlations. These trade-offs—shaped by temperature, diameter, and spatial layout—highlight design principles for neural architectures where coordination and information fidelity must be jointly optimized. The results establish that neural information flow is bounded not only by classical noise but also by irreducible geometric constraints, revealing a multi-dimensional optimization landscape with implications for biological and artificial systems.

Keywords: ephaptic coupling, mutual information, neural networks, geometric constraints, synchronization, information theory

Key Insight

Neural information transmission is shaped not only by classical noise sources but also by geometric constraints intrinsic to axonal architecture. This paper reveals that ephaptic coupling strength and axonal inclination jointly modulate a trade-off between synchronization and mutual information. While classical noise introduces stochastic variability, geometric misalignment—particularly beyond the 45-degree inclination threshold—leads to systematic breakdowns in coordinated signaling. Moreover, the manifold hypothesis suggests that maximal mutual information is bounded by high-dimensional geometric constraints in neural codespace, mirroring the limiting effects imposed by physical 3D geometry. Optimizing neural architectures requires balancing dynamic noise management with structural design to preserve fidelity and coordination.

1 Introduction

The transmission of information through neural networks represents one of the most fundamental processes in biological systems, yet the precise mechanisms governing information flow remain incompletely understood. Recent advances in the mathematical analysis of ephaptic coupling have revealed a complex interplay between the geometric configuration of axon tracts and their information-theoretic properties. This relationship challenges traditional views of neural communication and suggests that the physical architecture of neural networks directly constrains their computational capabilities through mechanisms that extend far beyond simple conduction velocity considerations.

Ephaptic coupling, the phenomenon whereby axons influence each other through current-mediated interactions in the extracellular space rather than through direct synaptic connections, has emerged as a critical factor in understanding lateral information transfer within neural networks. Unlike synaptic transmission, which represents discrete, directed communication events, ephaptic coupling creates continuous, bidirectional influences that depend critically on the spatial arrangement and electrical properties of the constituent axons. This

coupling mechanism becomes particularly significant in densely packed axon tracts where the proximity of neural fibers creates substantial opportunities for cross-talk and mutual influence.

The broader significance of this work lies in its demonstration that information-theoretic principles provide powerful tools for understanding neural system design and function. By quantifying the trade-offs between geometric constraints, coupling strength, and information capacity, this analysis reveals optimization principles that may guide both natural evolution and artificial neural network design. The mathematical framework developed here suggests that neural systems have evolved to navigate complex multi-dimensional optimization landscapes where physical, geometric, and information-theoretic constraints interact in subtle but fundamental ways.

The work demonstrates that neural information processing cannot be understood solely through traditional electrophysiological or computational approaches, but requires integrated analysis that considers geometric, physical, and information-theoretic constraints simultaneously. This interdisciplinary perspective reveals emergent properties of neural networks that arise from the complex interplay between their physical architecture and their information-processing requirements, providing new insights into both the fundamental principles governing neural computation and the practical constraints that shape neural system design and evolution.

2 Methods

2.1 Preliminaries

To rigorously assess ephaptic coupling dynamics, we simulated action potential propagation using a modified Hodgkin-Huxley framework tailored for tract geometries (see the adjacent figure which is due to Chawla-Morgera-Snider 2019). Building on the cable model, the simulations incorporated angular inclination θ and a phenomenological interaxonal distance ma-

trix W_{ip} to capture spatial dependencies. The extracellular region was assumed isopotential along transverse planes, enabling us to define $\phi(r, t) = \phi_{ext}(z, t)$ and derive regional formulations for intra- and extracellular potentials. Ion currents followed Frankenhauser-Huxley kinetics at nodes of Ranvier, with the governing equation extended to include ephaptic terms modulated by tract geometry. Simulations were performed using MATLAB, with axonal diameters, membrane capacitance, and conductivity computed per physical tract configuration. Time-domain solutions of the generalized coupling equation (Eq. 10 of Chawla-Morgera-Snider 2019) were obtained by finite difference discretization. To highlight the influence of geometry on synchrony and spike initiation timing, we varied axonal inclination and W_{ip} systematically across simulation sets. Injuries were modeled by altering inclination while preserving mean interaxonal distances. Synchronization studies employed pairs of axons with unequal diameters to assess velocity matching. All simulations used a tract length of 6–8 cm and time steps on the order of microseconds. Results were validated by comparing uncoupled and coupled cases for subthreshold and suprathreshold responses across configurations.

2.2 Geometric Relativity Framework

The geometric relativity of action potentials introduces constraints that fundamentally limit the configurations under which synchronization between axons can occur. Mathematical analysis demonstrates that synchronization is achievable only when axon inclinations relative to the tract axis remain below approximately forty-five degrees. This geometric threshold emerges from the mathematical equivalence between universal and typical coordinate models, where the universal coordinate $z_U = s_i \cos \theta_i$ captures individual axon projections while the typical coordinate $z_A = \sum_{i=1}^N s_i \cos \theta_i / N$ represents their average. The equivalence of these models within a constant integer factor reveals that synchronization conditions depend critically on the geometric relationship between individual axon orientations and the overall tract architecture.

When axons exceed this geometric threshold, the projection discrepancies become suffi-

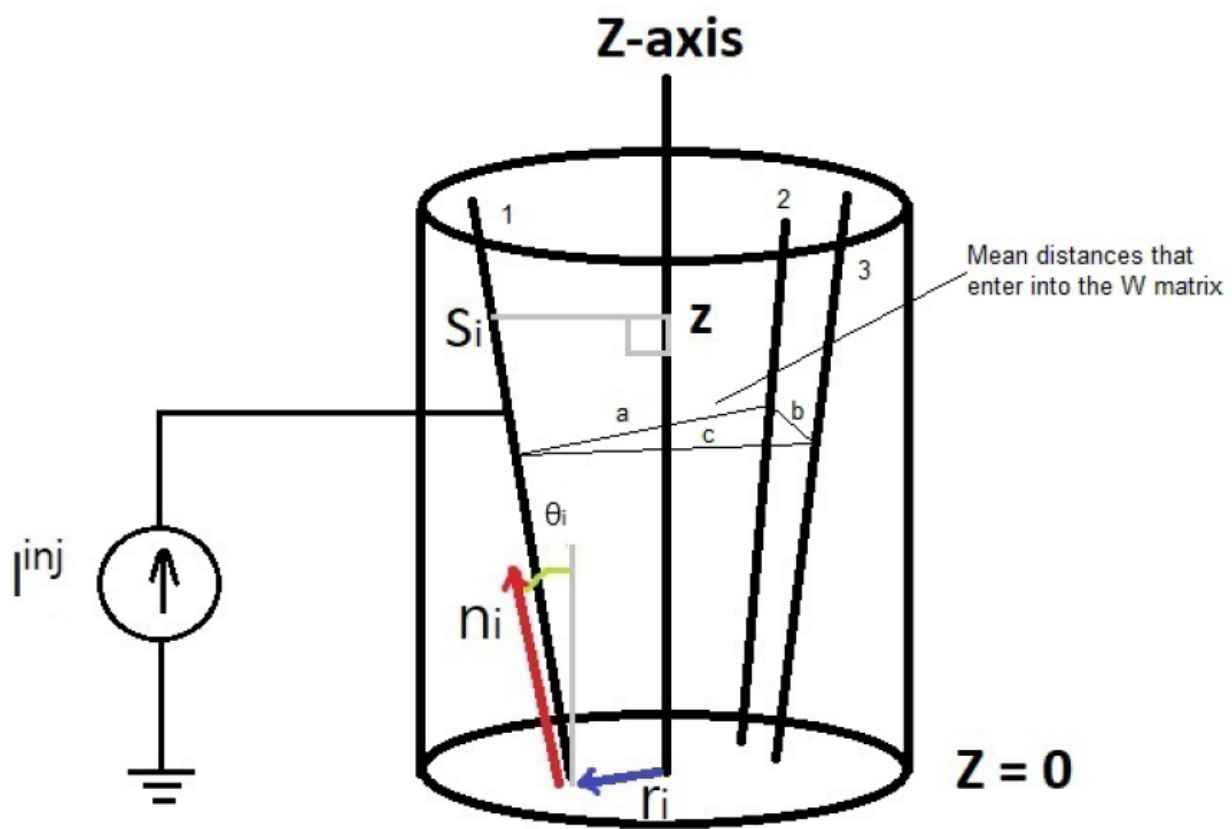


Figure 1: Axon bundle showing key parameters. Credits: Chawla-Morgera-Snider-2019.

ciently large that the mathematical framework underlying ephaptic synchronization breaks down. This breakdown is not merely a theoretical limitation but represents a fundamental physical constraint imposed by the geometry of three-dimensional space and the wave-like propagation of electrical signals along neural fibers. The geometric restriction thus defines distinct operational regimes for neural tracts, separating configurations that support coordinated activity from those that maintain independent information channels.

2.3 Information-Theoretic Analysis

The mathematical framework for quantifying the relationship between coupling strength and information transmission begins with the representation of voltage signals in terms of complete orthonormal basis functions, typically Fourier components, which transform temporal waveforms into discrete coefficient vectors. The mutual information between input coefficients $\{v_{0i}\}$ and output coefficients $\{v_{Li}\}$ can then be computed using empirical probability mass functions derived from numerical simulations. The resulting mutual information $I(v_{0j}; v_{Lj}) = H(v_{0j}) - H(v_{0j}|v_{Lj})$ quantifies the information preserved through the transmission process, where entropy terms capture the uncertainty in signal distributions.

The mathematical framework developed for analyzing these relationships extends beyond simple pairwise interactions to encompass the full complexity of multi-axon systems. The coupling matrix W that characterizes ephaptic interactions becomes a matrix of vectors rather than scalars when extracellular anisotropy is considered, reflecting the directional preferences in conductance that arise from the non-isotropic nature of neural tissue. This enhanced model captures the reality that ephaptic coupling depends not only on the distance between axons but also on their relative orientations and the properties of the intervening extracellular medium.

2.4 Hodgkin-Huxley Model Implementation

The information-theoretic analysis of coupled axon systems applies mutual information theory to the Hodgkin-Huxley equations governing axonal dynamics. This approach allows for rigorous quantification of how ephaptic coupling affects signal transmission by modeling the voltage dynamics of individual axons while accounting for their mutual electrical influences through the extracellular medium.

3 Results

3.1 Geometric Constraints on Synchronization

The analysis reveals that synchronization between axons is achievable only when axon inclinations relative to the tract axis remain below approximately 45 degrees. This geometric threshold represents a fundamental physical constraint that separates neural tract configurations into distinct operational regimes: those supporting coordinated activity and those maintaining independent information channels.

3.2 Coupling Strength Effects on Information Capacity

The information-theoretic analysis reveals a counterintuitive relationship between ephaptic coupling strength and information transmission capacity. Increased coupling strength systematically reduces the mutual information between input and output signals. This reduction occurs because ephaptic coupling introduces correlations between previously independent channels, effectively reducing the dimensionality of the information space available for signal transmission.

Numerical experiments demonstrate that mutual information exhibits a complex, approximately convex relationship with coupling strength, initially decreasing as coupling increases before showing slight recovery at very high coupling levels. This behavior suggests that

ephaptic coupling acts as a form of noise or distortion from an information-theoretic perspective, despite its role in enabling synchronization and coordinated neural activity. The apparent contradiction between the coordinating effects of coupling and its detrimental impact on information capacity reveals a fundamental trade-off in neural system design.

3.3 Joint Geometric-Information Relationship

The joint impact of geometry and coupling strength on maximal mutual information emerges through the intersection of these two constraint mechanisms. Within the geometrically permissible regime where axon inclinations allow synchronization, the strength of ephaptic coupling determines the extent to which information capacity is degraded. Conversely, in geometric configurations that exceed the synchronization threshold, ephaptic coupling may have different effects on information transmission, potentially preserving individual channel capacities while preventing coordinated activity.

This geometric-information relationship has profound implications for understanding neural network architecture and evolution. Neural systems appear to face an optimization problem that balances the benefits of coordinated activity against the costs of reduced information transmission capacity. In contexts where precise information transmission is paramount, such as sensory pathways requiring high fidelity signal reproduction, neural architectures may favor geometric configurations that minimize ephaptic coupling while maintaining adequate conduction properties. Conversely, in systems where coordination and synchronization are essential, such as motor control networks or oscillatory circuits involved in cognitive processes, the architecture may accept reduced information capacity in exchange for enhanced synchronization capabilities.

3.4 Temperature and Diameter Effects

Temperature effects introduce additional complexity to the geometric-information relationship. The analysis reveals that mutual information exhibits a non-monotonic relationship

with temperature, initially increasing before declining at higher temperatures. This behavior likely reflects the competing effects of improved ion channel kinetics at moderate temperatures versus the degradation of signal fidelity at excessive temperatures where conduction becomes too rapid for proper signal integration. The temperature dependence suggests that thermal regulation may serve not only metabolic functions but also information-theoretic optimization in neural systems.

Axon diameter represents another critical parameter influencing the geometric-information trade-off. Larger diameter axons typically conduct signals more rapidly and with greater reliability, yet the analysis suggests that increased diameter may actually reduce mutual information in the context of ephaptic coupling. This counterintuitive result likely reflects the increased coupling strength that accompanies larger axon cross-sections, leading to greater correlation between channels and reduced effective information capacity. The diameter effect thus represents another dimension of the optimization problem faced by neural architectures.

3.5 Pathological and Clinical Implications

The implications of these findings extend to pathological conditions affecting neural tissue. Diseases that alter axon geometry, such as those causing demyelination or structural disruption, may fundamentally change the geometric-information relationship within affected tracts. Similarly, conditions that modify extracellular conductivity or ion concentrations may shift the coupling strength parameters, potentially moving neural circuits between different operational regimes with distinct information-processing characteristics.

3.6 Future Directions

Future research directions suggested by this analysis include the extension to axon systems with branching as well as tortuosity, the incorporation of dynamic geometric changes that might occur during development or plasticity, and the investigation of how these principles apply to other forms of neural coupling beyond ephaptic interactions. The mathematical

tools developed here may also prove applicable to understanding information flow in other biological networks where geometric constraints and coupling strength jointly determine functional properties. Even more importantly, since lack of experimental validation is one of the limitations of our work, this study calls for detailed neurophysiological experiments for validation, involving rodent models that examine nerve tracts either in the limbs or perhaps even the corpus callosum for joint geometric and ephaptic contributions to throughput in the mammalian nervous system.

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Appendix: Limits When Maximizing the Input-Output Mutual Information of Nerve Tracts

The analysis of ephaptic coupling and geometric constraints in nerve tracts reveals fundamental information-theoretic limits that extend beyond traditional stochastic noise models. Recent work on the manifold hypothesis (MH) provides a complementary perspective on these limits, suggesting that even in the absence of classical channel noise, irreducible geometric distortions impose a minimal noise floor on neural information transmission.

The geometric constraints governing axon tract synchronization—specifically the requirement that axon inclinations remain below 45 degrees—can be understood as manifestations of the manifold-induced distortion described in MH theory. Neural signals propagating along axon tracts exist within a high-dimensional state space that is effectively constrained to a lower-dimensional manifold defined by the biophysical properties of the Hodgkin-Huxley equations and the geometric configuration of the tract. When axons exceed the critical inclination threshold, the projection discrepancies between individual axon coordinates and the

average tract coordinate create geometric mismatches analogous to the quantization errors that occur when uniform grids are applied to curved manifolds.

This geometric interpretation provides new insight into why ephaptic coupling systematically reduces mutual information between input and output signals. The coupling mechanism introduces correlations that effectively reduce the dimensionality of the information space, similar to how manifold constraints limit the effective degrees of freedom available for signal representation. The observed convex relationship between coupling strength and mutual information—where information capacity initially decreases before showing slight recovery at high coupling levels—parallels the trade-offs inherent in manifold-aware encoding schemes, where improved local fidelity along the manifold comes at the cost of increased vulnerability to orthogonal perturbations.

The analogy to quantum nondemolition (QND) measurements is particularly illuminating. Just as QND measurements with squeezed states can reduce uncertainty in one observable only by increasing uncertainty in conjugate variables, ephaptic coupling can enhance synchronization and coordination only by reducing the independence of information channels. This represents a fundamental constraint that cannot be eliminated through engineering improvements but must be managed through architectural optimization.

The temperature and diameter effects observed in nerve tract analysis further support this geometric interpretation. The non-monotonic relationship between temperature and mutual information reflects the competing demands of improved ion channel kinetics versus signal integration fidelity—a trade-off that emerges naturally from the geometric constraints of the underlying neural manifold. Similarly, the counterintuitive reduction in mutual information with increased axon diameter reflects the geometric reality that larger cross-sections enhance coupling strength and thus increase channel correlations.

These insights suggest that neural architectures have evolved to navigate a multi-dimensional optimization landscape where geometric, physical, and information-theoretic constraints interact in ways that parallel the fundamental limits described by MH theory. The 45-degree

inclination threshold represents not merely a synchronization condition but a manifestation of deeper geometric principles that govern information flow in spatially distributed systems.

Future research should investigate whether adaptive coding strategies inspired by manifold-aware learning—such as those employed in variational autoencoders and diffusion models—might provide new approaches for optimizing information transmission in neural networks while respecting geometric constraints. The recognition that neural information processing faces irreducible geometric limits, rather than merely stochastic ones, opens new avenues for understanding both biological neural architectures and artificial network design principles.

Manifold Framework Applied to Ephaptic Coupling

Applying the manifold framework directly to ephaptic coupling reveals that neural signals in coupled axon tracts exist on a constraint manifold $\mathcal{M}$ embedded in the high-dimensional voltage space $\mathbb{R}^N$, where N represents the number of spatial and temporal sampling points. The Hodgkin-Huxley dynamics, combined with geometric coupling constraints, define a smooth d -dimensional manifold where $d \ll N$. Traditional uniform quantization of voltage signals—as typically employed in digital neural interfaces or computational models—inevitably produces distortion errors orthogonal to $\mathcal{M}$.

The 45-degree geometric threshold can be reinterpreted as the critical curvature beyond which uniform encoding schemes fail catastrophically. When axon inclinations exceed this limit, the local tangent spaces $T_x\mathcal{M}$ vary so rapidly that no fixed quantization grid can maintain fidelity across the tract. This geometric insight explains why synchronization breaks down: the manifold curvature induced by extreme inclinations makes uniform signal representation impossible, forcing individual axons into independent operational regimes.

Ephaptic coupling strength directly modulates the manifold’s intrinsic dimensionality. Weak coupling preserves the independence of axonal channels, maintaining high effective dimensionality, while strong coupling constrains signals to lower-dimensional submanifolds

characterized by synchronized dynamics. The observed reduction in mutual information with increased coupling strength reflects this dimensional collapse, where the constraint manifold becomes increasingly narrow, limiting the available degrees of freedom for independent information encoding. This geometric perspective suggests that optimal neural architectures must balance manifold dimensionality against coordination requirements.

Summary

The failure of uniform quantization on curved manifolds reflects a geometric constraint that parallels the breakdown of synchronization in axon tracts exceeding the critical 45-degree inclination. At such inclinations, projection mismatches disrupt the shared coordinate structure required for coordinated activity, resulting in independent signal regimes. This transition mirrors the loss of encoding fidelity observed in manifold curvature beyond which uniform representation becomes untenable. Ephaptic coupling cannot reestablish synchronization when geometric divergence surpasses representational thresholds. Accordingly, the synchronization threshold is not merely a dynamic constraint but a manifestation of deeper geometric encoding limits—a convergence point between spatial architecture and information-theoretic fidelity in neural transmission.

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